FMAN45-Assignment 1

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We start by solving equation 1 in the given instruction:

$$\begin{aligned} &\frac{1}{2}\left\|\mathbf{r_i} - \mathbf{x_i} w_i\right\|_2^2 + \lambda \left\|w_i\right\| \\ &= \frac{1}{2}(\mathbf{r_i}^T\mathbf{r_i} - 2w_i\mathbf{x_i}^T\mathbf{r_i} + w_i^2\mathbf{x_i}^T\mathbf{x_i}) + \lambda \left\|w_i\right\| \end{aligned} \tag{1}$$

If we derive this w.r.t w_i we get:

$$-\mathbf{x_i}^T\mathbf{r_i} + w_i\mathbf{x_i}^T\mathbf{x_i} + \mathbf{sign}(w_i)\lambda \tag{2}$$

To minimize, we set to zero and solve for w_i:

$$\begin{aligned} w_i &= \frac{\mathbf{x_i}^T\mathbf{r_i} + \lambda}{\mathbf{x_i}^T\mathbf{x_i}} \text{ if } w_i < 0\text{; valid when: } \mathbf{x_i}^T\mathbf{r_i} < -\lambda \\ w_i &= \frac{\mathbf{x_i}^T\mathbf{r_i} - \lambda}{\mathbf{x_i}^T\mathbf{x_i}} \text{ if } w_i > 0\text{; valid when } \mathbf{x_i}^T\mathbf{r_i} > \lambda \end{aligned} \tag{3}$$

We can rewrite this and analyze the sign of $\mathbf{x_i}^T\mathbf{r_i}$ as:

$$\begin{aligned} w_i &= \frac{\mathbf{x_i}^T\mathbf{r_i}}{\mathbf{x_i}^T\mathbf{x_i}}(1 + \frac{\lambda}{\mathbf{x_i}^T\mathbf{r_i}}) = \frac{\mathbf{x_i}^T\mathbf{r_i}}{\mathbf{x_i}^T\mathbf{x_i}}(1 - \frac{\lambda}{|\mathbf{x_i}^T\mathbf{r_i}|}) \text{ if } w_i < 0\text{; valid when } \mathbf{x_i}^T\mathbf{r_i} < -\lambda \\ w_i &= \frac{\mathbf{x_i}^T\mathbf{r_i}}{\mathbf{x_i}^T\mathbf{x_i}}(1 - \frac{\lambda}{\mathbf{x_i}^T\mathbf{r_i}}) = \frac{\mathbf{x_i}^T\mathbf{r_i}}{\mathbf{x_i}^T\mathbf{x_i}}(1 - \frac{\lambda}{|\mathbf{x_i}^T\mathbf{r_i}|}) \text{ if } w_i > 0\text{; valid when } \mathbf{x_i}^T\mathbf{r_i} > \lambda \end{aligned} \tag{4}$$

And we get the answer:

$$\hat{w_i} = \frac{\mathbf{x_i}^T\mathbf{r_i}}{\mathbf{x_i}^T\mathbf{x_i}\,|\mathbf{x_i}^T\mathbf{r_i}|}\left(|\mathbf{x_i}^T\mathbf{r_i}| - \lambda\right) \quad \text{if } |\mathbf{x_i}^T\mathbf{r_i}| > \lambda \tag{5}$$

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Since we have an orthogonal matrix, we know that $\mathbf{x_i}^T\mathbf{x_i} = 1$. In equation 6 we use this identity and the fact that all other instances will be zero in the matrix

multiplication between $\mathbf{x_i}^T\mathbf{x_l}$:

$$\mathbf{x_i}^T\mathbf{r_i}^{j-1} = \mathbf{x_i}^T(\mathbf{t} - \sum_{l<i}\mathbf{x_l}\hat{w}_l^j - \sum_{l>i}\mathbf{x_l}\hat{w}_l^{j-1}) = \mathbf{x_i}^T\mathbf{t} \tag{6}$$

We use this expression for $\mathbf{x_i}^T\mathbf{r_i}^{j-1}$, the fact that: $\mathbf{x}_i^T\mathbf{x_i} = 1$ and equation 5 when calculating the difference:

$$\hat{w}_i^{j+1} - \hat{w}_i^j = \frac{\mathbf{x_i}^T\mathbf{t}}{|\mathbf{x_i}^T\mathbf{t}|}(|\mathbf{x_i}^T\mathbf{t}| - \lambda) - \frac{\mathbf{x_i}^T\mathbf{t}}{|\mathbf{x_i}^T\mathbf{t}|}(|\mathbf{x_i}^T|\mathbf{t} - \lambda) = 0 \tag{7}$$

And if we observe $\hat{w}_i$ for the other condition: $|\mathbf{x_i}^T\mathbf{r_i}| \leq \lambda$ the trivial solution becomes 0, because $\hat{w}_i = 0$ for all i and j.

3

If we start from the two equations in 3, when $w_i < 0$ and $w_i > 0$, and using the orthogonality discussed above, we get:

$$\begin{aligned}
\hat{w}_i^1 &= \frac{\mathbf{x_i}^T\mathbf{r_i} + \lambda}{\mathbf{x_i}^T\mathbf{x_i}} = \mathbf{x_i}^T\mathbf{t} + \lambda : \text{ valid when } \mathbf{x_i}^T\mathbf{t} < -\lambda \\
\hat{w}_i^1 &= \frac{\mathbf{x_i}^T\mathbf{r_i} - \lambda}{\mathbf{x_i}^T\mathbf{x_i}} = \mathbf{x_i}^T\mathbf{t} - \lambda : \text{ valid when } \mathbf{x_i}^T\mathbf{t} > \lambda
\end{aligned} \tag{8}$$

We then use the definition of $\mathbf{t}$ given in the instructions:

$$\begin{aligned}
\hat{w}_i^1 &= \mathbf{x_i}^T\mathbf{X}\mathbf{w}^* + \mathbf{x_i}^T\mathbf{e} + \lambda : \text{ valid when } \mathbf{x_i}^T\mathbf{X}\mathbf{w}^* + \mathbf{x_i}^T\mathbf{e} < -\lambda \\
\hat{w}_i^1 &= \mathbf{x_i}^T\mathbf{X}\mathbf{w}^* + \mathbf{x_i}^T\mathbf{e} - \lambda : \text{ valid when } \mathbf{x_i}^T\mathbf{X}\mathbf{w}^* + \mathbf{x_i}^T\mathbf{e} > \lambda
\end{aligned} \tag{9}$$

Furthermore, we have $\mathbf{x}_i^T \in R^{1\times n}$, $\mathbf{X} \in R^{n\times d}$, and $\mathbf{w}^* \in R^{d\times 1}$. Assuming orthonormal columns of $\mathbf{X}$, it follows that $\mathbf{x}_i^T\mathbf{x}_i = 1$. Hence, we obtain:

$$\begin{aligned}
\hat{w}_i^1 &= w_i^* + \mathbf{x_i}^T\mathbf{e} + \lambda \text{ :valid when } w_i^* + \mathbf{x_i}^T\mathbf{e} < -\lambda \\
\hat{w}_i^1 &= w_i^* + \mathbf{x_i}^T\mathbf{e} - \lambda \text{ :valid when } w_i^* + \mathbf{x_i}^T\mathbf{e} > \lambda
\end{aligned} \tag{10}$$

We can then calculate the expectation value by:

$$\begin{aligned}
&\lim_{\sigma\to 0} E[\hat{w_i}^1 - w_i^*] = \lim_{\sigma\to 0} E[w_i^* + \mathbf{x_i}^T\mathbf{e} - w_i^* + \lambda] = \lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e} + \lambda] \\
&: \text{ valid when } w_i^* + \lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e}] < -\lambda \\
&\lim_{\sigma\to 0} E[\hat{w_i}^1 - w_i^*] = \lim_{\sigma\to 0} E[w_i^* + \mathbf{x_i}^T\mathbf{e} - w_i^* - \lambda] = \lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e} - \lambda] \\
&: \text{ valid when } w_i^* + \lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e}] > \lambda
\end{aligned} \tag{11}$$

Since $\mathbf{e} \to 0$ as $\sigma \to 0$, due to the Gaussian distribution, we can write the equations above (11) as:

$$\begin{aligned}
&\lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e} + \lambda] = \mathbf{x_i}^T \lim_{\sigma\to 0} E[\mathbf{e}] + \lim_{\sigma\to 0} E[\lambda] \\
&: \text{valid when } w_i^* + \lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e}] < -\lambda \\
&= \lim_{\sigma\to 0} E[\lambda] : \text{valid when } w_i^* < -\lambda \\
&= \lambda : \text{valid when } w_i^* < -\lambda \\
&\lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e} - \lambda] = \mathbf{x_i}^T \lim_{\sigma\to 0} E[\mathbf{e}] - \lim_{\sigma\to 0} E[\lambda] \\
&: \text{valid when } w_i^* + \lim_{\sigma\to 0} E[\mathbf{x_i}^T\mathbf{e}] > \lambda \\
&= \lim_{\sigma\to 0} -E[\lambda] : \text{valid when } w_i^* > \lambda \\
&= -\lambda : \text{valid when } w_i^* > \lambda
\end{aligned} \tag{12}$$

Finally, we consider the case where $|\mathbf{x_i}^T\mathbf{r_i}| \leq \lambda$ i.e $\hat{w}_i^1 = 0$. Note that from the above we have proven that : $\mathbf{x_i}^T\mathbf{r_i} \to \hat{w}_i^*$ as $\sigma \to 0$

$$\begin{aligned}
&\lim_{\sigma\to 0} E[\hat{w_i}^1 - w_i^*] = \lim_{\sigma\to 0} E[0 - w_i^*] \\
&= -E[w_i^*] = -w_i^* : \text{valid when } |\mathbf{x_i}^T\mathbf{r_i}| \leq \lambda \text{ i.e } |w_i^*| \leq \lambda
\end{aligned} \tag{13}$$

We can summarize the results above:

$$\lim_{\sigma\to 0} E\left(\hat{w}_i^{(1)} - w_i^*\right) = \begin{cases} -\lambda, & w_i^* > \lambda \\ -w_i^*, & |w_i^*| \leq \lambda \\ \lambda, & w_i^* < -\lambda \end{cases} \quad \forall i \tag{14}$$

The LASSO method is closely related to its acronym. It uses the *Least* squares loss because it tries to fit the data by minimizing $\|\mathbf{t} - \mathbf{X}\mathbf{w}\|_2^2$. Absolute comes from the fact that the penalty uses the *absolute* value of the weights, $||\mathbf{w}_|||_1$, rather than squared. This penalty-term *shrinks* the weights toward zero. Since some of the weights shrink exactly to zero, LASSO performs a *selection* of features.

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From Figures 1 - 3 and Table 1 the problem of overfitting and under-fitting can be addressed. As one can observe, when the lambda is smaller, the penalty is smaller and more weights are non-zero. The model captures noise, high variance, and low bias. As can be seen, Figure 1 oscillates more than the others. Hence, the model is over fitted. When λ increases, the LASSO regularization penalizes harder and more weights are set to zero. This makes the model smoother, but fails to capture the true structure of the data. The model is then under-fitted. We want to find the value of λ that captures the main trend of the data without

fitting to noise. Since the signal contains two frequencies, and each frequency uses a base of sine and cosine, we need at least 4 weights to be able to represent the signal. Because the signal contains noise, we need more weights than 4 to represent the signal.

	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$
Non-zero	238	63	6

Table 1: number of non-zero coordinates for different values of λ.

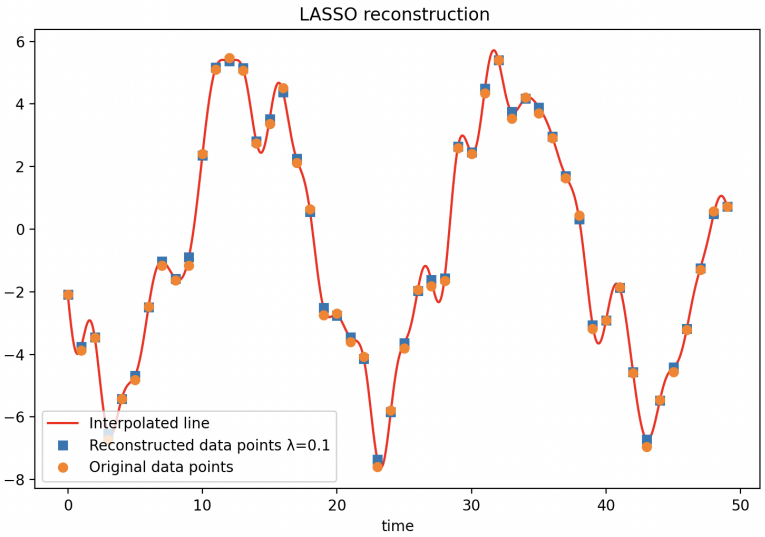


Figure 1: LASSO reconstruction for $\lambda = 0.1$

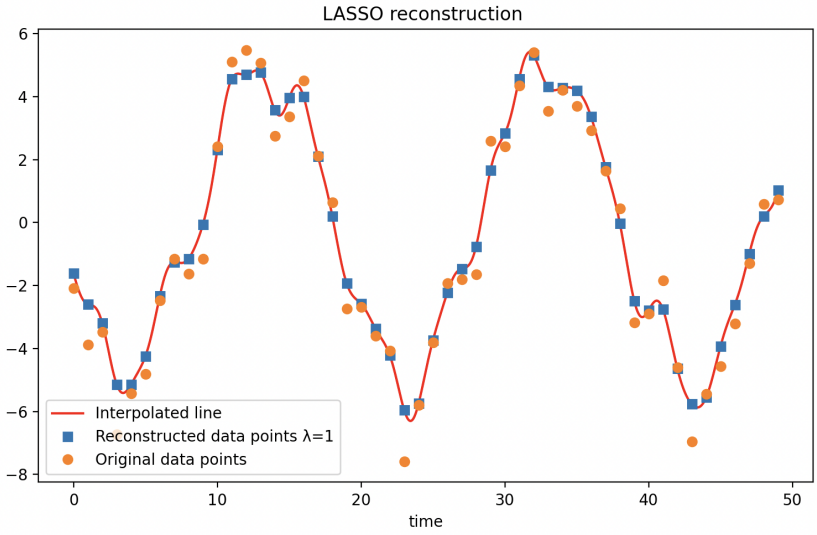


Figure 2: LASSO reconstruction for $\lambda = 1$

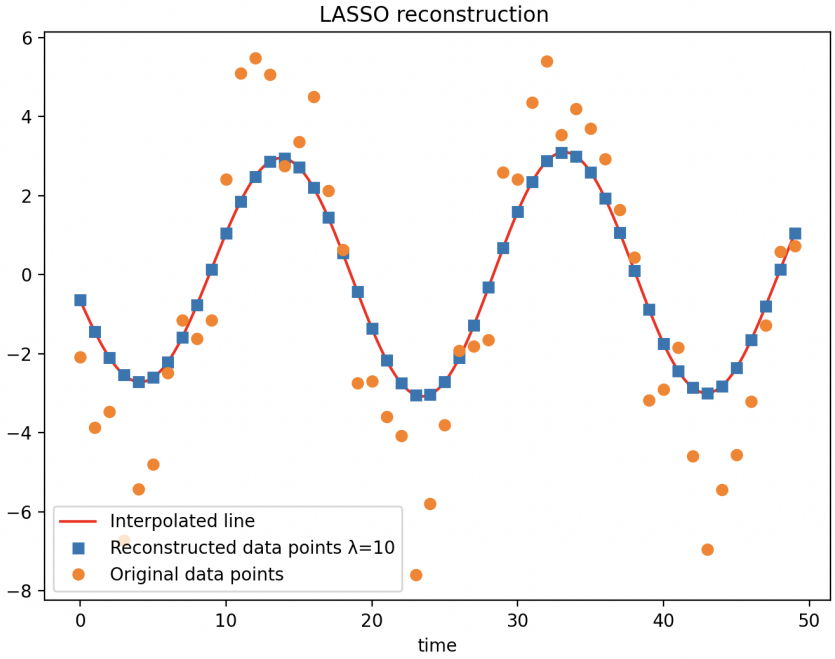


Figure 3: LASSO reconstruction for $\lambda = 10$

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In Figure 4 we see how $RMSE_{val}$ explains the problem of overfitting and underfitting. For smaller λ the validation error is large, due to overfitting of the training data, therefore, we have a small $RMSE_{est}$ value. When λ increases, $RMSE_{val}$ decreases due to better generalization because we reduce the number of weights. When λ increases even more, $RMSE_{val}$ will increase due to underfitting, we have too many zero valued weights and a to simple model, even to simple to fit the training data and the $RMSE_{est}$ increases. The minimum $RMSE_{val}$ value represent the perfect trade off between overfitting and underfitting. This λ provides the best generalization performance. In Figure 5 the generalization is visualized, a clear difference can be observed from the over-fitted model in Figure 1 and the under-fitted model in Figure 3.

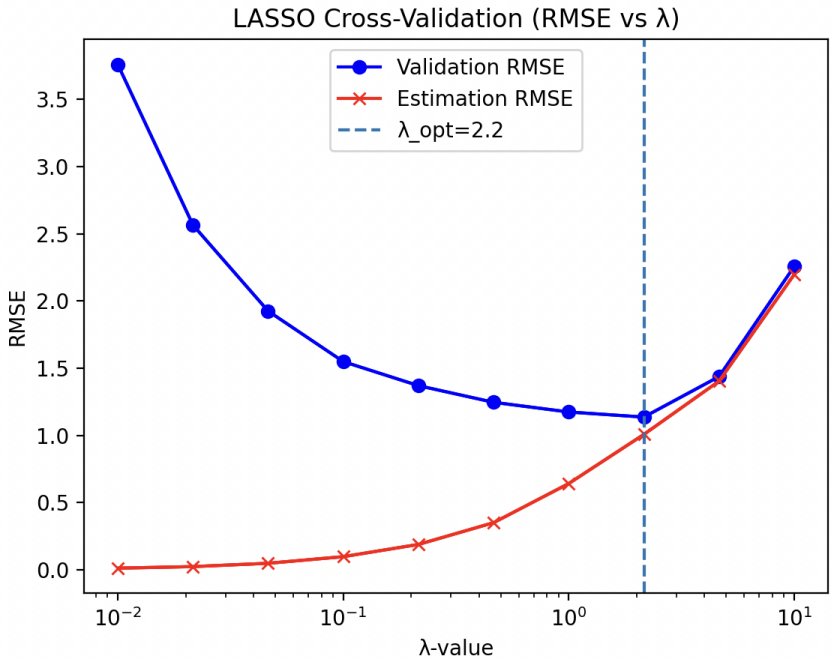


Figure 4: $RMSE_{val}$ and $RMSE_{est}$ as a function of different λ values

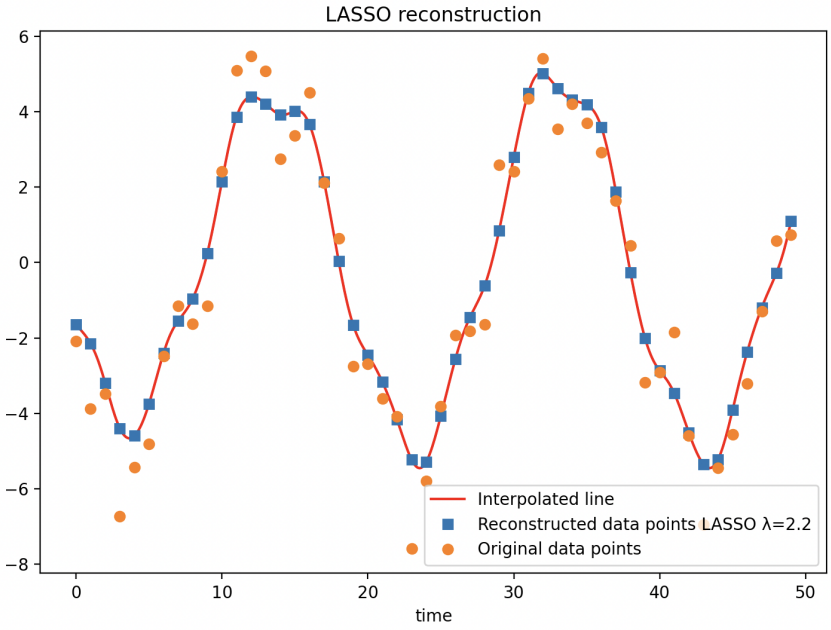


Figure 5: LASSO reconstruction for the optimal $\lambda = 2.2$

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If we compare Figure 4 and Figure 6 we can observe the same general behavior, the U-shaped $RMSE_{val}$ curve and the increasing $RMSE_{est}$ curve. The same reasoning holds for the behavior of the errors as stated above. However, since cross validation is performed over different frames, for small λ the model overfits individual frames and captures noise in the data, the reconstruction becomes unstable. For larger λ the model is theoretically too simple to capture the underlying music structure. Both $RMSE_{est}$ and $RMSE_{val}$ flatten out when λ is large because the model has reached the simplest possible state, most weights are set to zero.

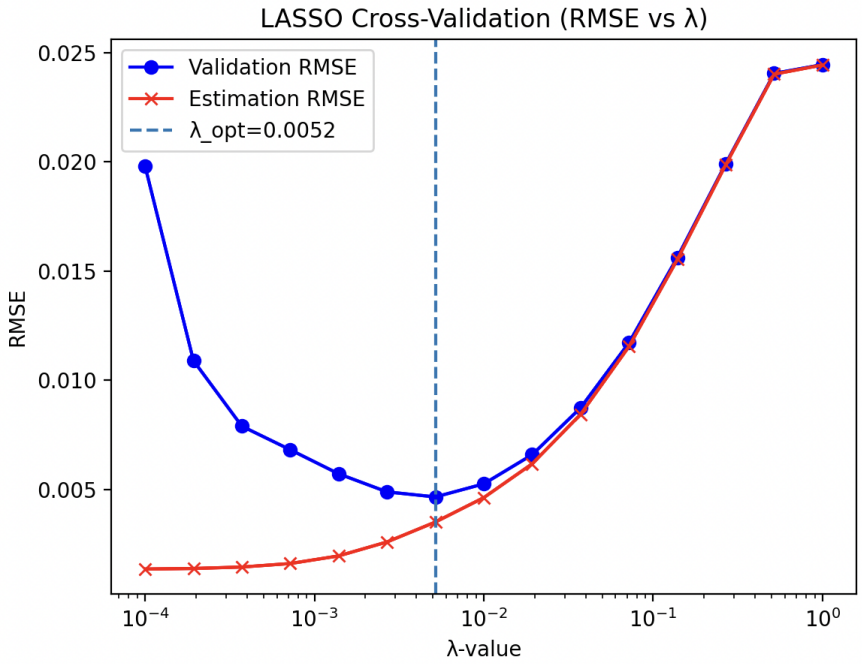


Figure 6: $RMSE_{val}$ and $RMSE_{est}$ as a function of different λ values

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There is a clear difference in the perception of sound for different values of λ. When having a smaller λ the noise is preserved and the output is less clean. When the value of λ increases, regularization and penalization increase, leading to less noise but also to the removal of the underlying musical structure and to a more smooth sound. When listening to reconstructed audio with the optimal λ a reduction in background noise is noticeable. However, this λ provides optimal

performance in terms of RMSE, not perceived audio quality. Maybe human audio perception is not proportional to the RMSE error. Hence, the best result is the trade off between the objective minimization of the error and the subjected sound quality.